

**ABV-Indian Institute of Information Technology and Management (ABV-IIITM),
Gwalior**

Department of Information Technology

Software Engineering Laboratory

Lab Sheet – 1

Date: 31 July 2026

Course: Software Engineering Laboratory

Experiment No.: 1

Title: Student Grade Processing System – Functional Testing

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Aim

To develop a Student Grade Processing System and validate its functionality using software testing.

Problem Statement

An Excel spreadsheet containing **10–20 student records** will be provided by the instructor.

The spreadsheet contains the following columns:

- Roll Number
- Student Name
- Department
- Subject
- Project Marks (20)
- Minor Marks (20)
- Laboratory Marks (20)
- End Examination Marks (40)
- Grade (*Empty*)

Total Marks = Project + Minor + Laboratory + End Examination (Maximum = 100)

Write a program that:

- Reads the uploaded spreadsheet.
 - Calculates the **Total Marks**.
 - Determines the student's **Grade**.
 - **Adds the Grade** to the existing **Grade** column.
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Grading Policy

Total Marks (Out of 100) Grade

≥ 90	A
75–89	B
60–74	C
35–59	D
< 35	F

Laboratory Tasks

Task 1

Download the Excel sheet uploaded by the instructor.

Task 2

Write a program to:

- Read the spreadsheet.
- Calculate the total marks.
- Assign the appropriate grade.
- Fill the **Grade** column in the spreadsheet.

Task 3

Execute the following **test cases** by adding **one new student record** for each test case.

Test Cases

Test Case 1 – Invalid Total Marks

Add a new student record such that the total marks become **101** (for example, Project = 20, Minor = 20, Laboratory = 20, End Examination = 41). Observe the system behavior and record the output. If the input is invalid, the program should display an appropriate error message and should not assign a grade.

Test Case 2 – Valid Student Record

Add a new student record with a **total of 60 marks**. Execute the program and observe the generated grade. Record the actual output and determine whether it matches the grading policy.

Test Case 3 – Missing Input

Add a new student record where any one of the marks (Project, Minor, Laboratory, or End Examination) is left **blank (NULL)**. Execute the program and observe how the system handles incomplete input. Record the displayed message or behavior.

Test Case 4 – Invalid Data Type (Tricky Test Case)

Add a new student record where one of the marks contains a **non-numeric value** (for example, "AB" or "Absent" instead of a numeric mark). Execute the program and observe the system behavior. Record the output and identify the type of error generated.

Instructions

- Use the uploaded Excel sheet as the input.
- For **valid test cases**, write the computed **Grade** into the **Grade** column.
- For **invalid test cases**, **do not assign a grade** and display an appropriate error message.
- Record the outcome of each test case as **PASS** or **FAIL** along with the reason.

Deliverables

1. Source code.
2. Input Excel sheet.
3. Updated Excel sheet with the **Grade** column filled for valid records.
4. Screenshot of program execution.

5. Test case execution report showing:

- Test Case ID
- Expected Result
- Actual Result
- PASS/FAIL
- Error Message (if applicable)

Expected Outcome

Students will develop a simple Student Grade Processing System and validate its correctness using functional test cases. The focus of this experiment is on **executing test cases, validating inputs, identifying errors, and ensuring that grades are assigned only for valid student records.**